Navigating Volatility: Federal Reserve Policy and Market Liquidity (1986-1990)

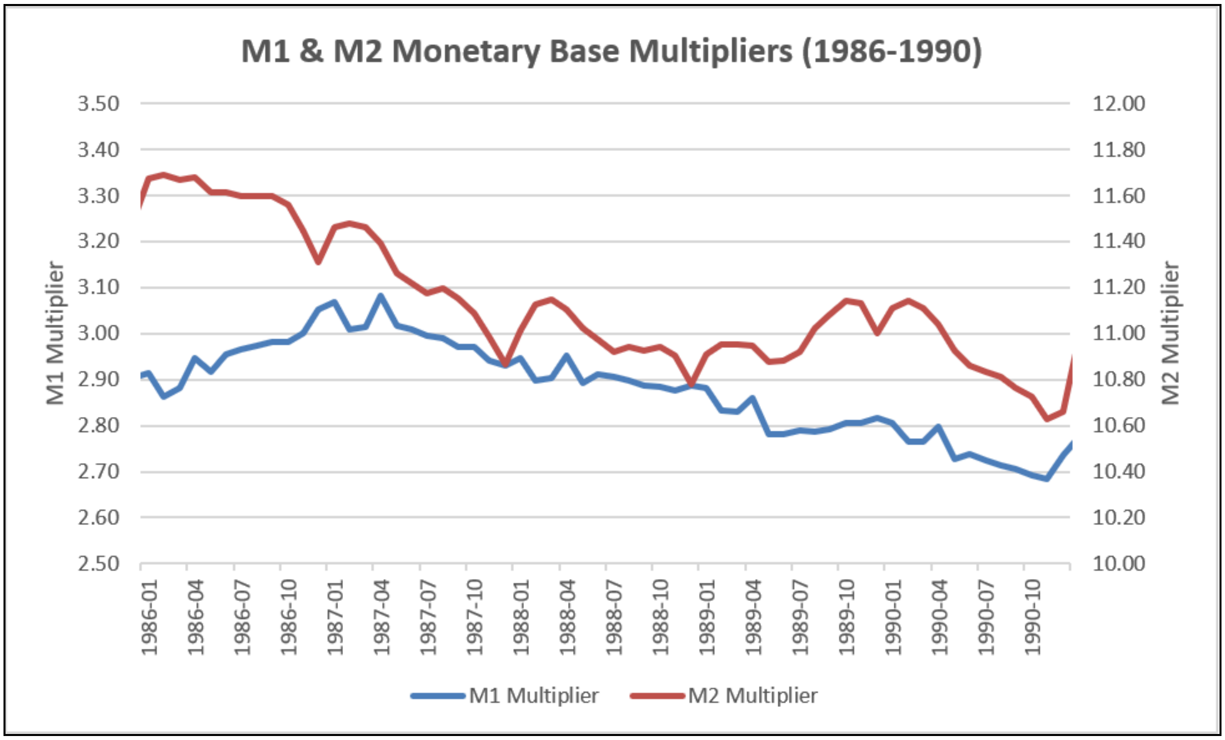
Introduction

On October 19, 1987, global markets cratered in what became known as Black Monday, forcing the Federal Reserve into a high-stakes balancing act between saving the financial system and triggering runaway inflation. Between 1986 and 1990, the Fed navigated this highly volatile environment that was a stock market crash followed immediately by an overheating economy. Throughout this period, the central bank's primary mandates were supporting long-term price stability and sustainable economic growth. To achieve these objectives, the Federal Reserve utilized open market operations (OMO) as its primary policy tool. By targeting the "degree of reserve pressure" (non-borrowed reserves) in the short term, the Fed sought to influence its intermediate targets of the M2 and M3 money supplies.

Money Supply Dynamics

During the late 1980s, financial deregulation and the introduction of interest-bearing checking accounts made M1highly unpredictable. As a result of this, the Federal Open Market Committee (FOMC) shifted its focus from strictly tracking M1 and instead targeting broader aggregates such as M2 and M3. As noted in the 1988 FOMC documents, the committee established a growth target range of 4 to 8 percent for M2.

From 1986 to 1990, the monetary base multipliers for both M1 and M2 experienced a steady decline, as illustrated below.



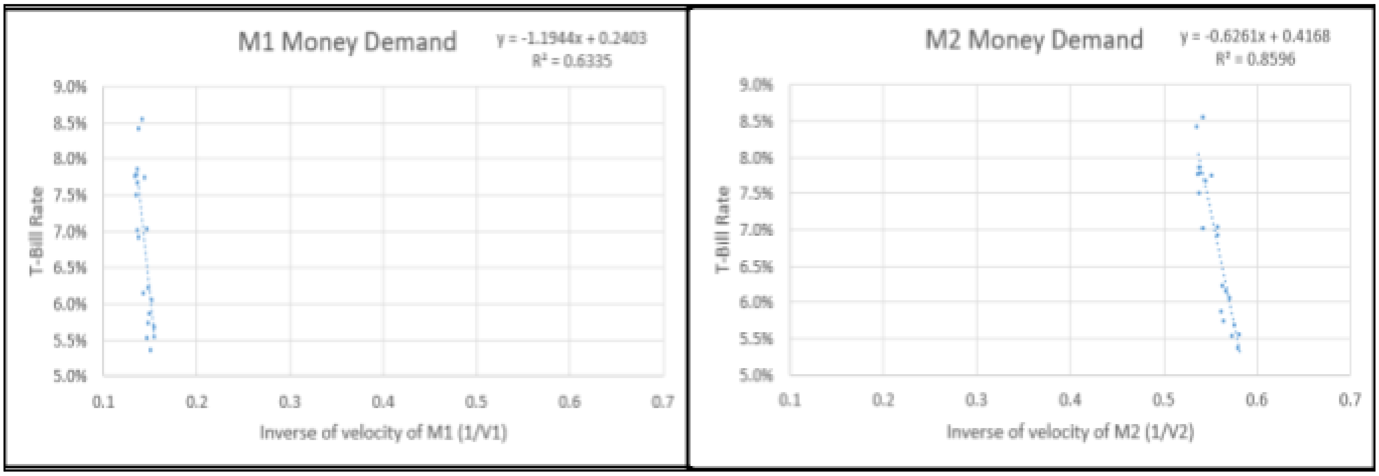
This contraction can be mathematically analyzed through the monetary base multiplier equation:

$$m = \frac{1 + C/D}{r_0 + ER/D + C/D}$$

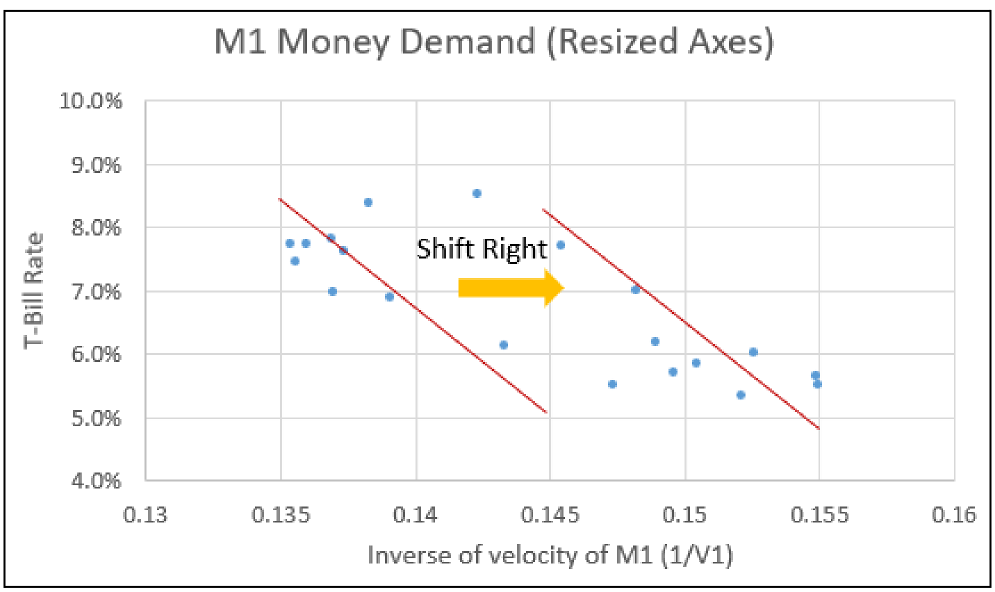
During this timeframe, the required reserve ratio (R_0) remained relatively constant. Banks did not accumulate excess reserves (ER/D). Therefore, the decline in the multiplier was primarily driven by an increase in the currency-to-deposit ratio (C/D). As the public held more physical currency relative to bank deposits, the banking sector's lending capacity decreased, slowing the money creation process and driving the multiplier downward.

Volatility in Money Demand

Analyzing the relationship between the T-bill rate and the inverse velocity (1/V) highlights the relative stability of money demand during this era. Plotting M1 and M2 money demand side-by-side reveals the expected downward slope. This indicates that as interest rates fall, the public holds more money.



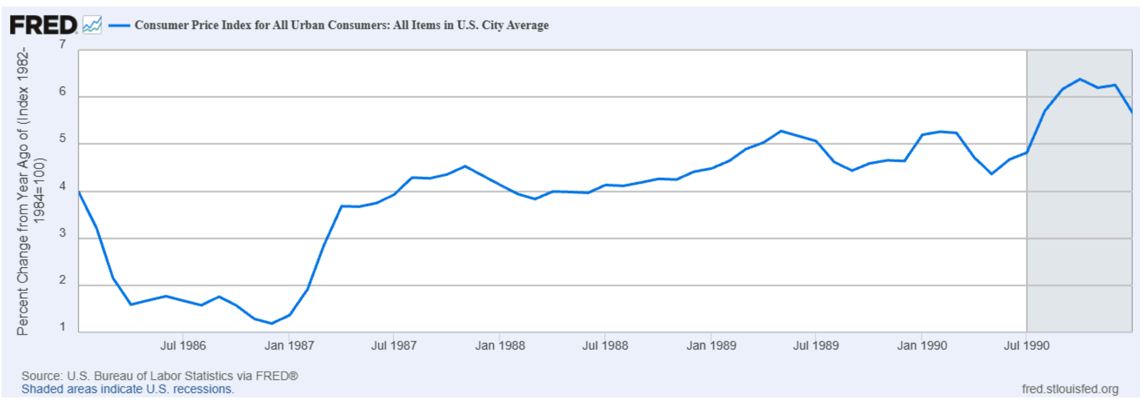
However, M2 demonstrated significant stability, plotting as a consistent and tighter trendline. M1 was more scattered and volatile in comparison. A closer examination of the resized M1 axes reveals a distinct rightward shift in demand around 1989.



During this shift, the demand for M1 increased suddenly despite minimal changes in interest rates. This fundamental instability illustrates exactly why the Federal Reserve abandoned M1 as a strict target. Enforcing a specific M1 supply during shifting demand would have resulted in severe interest rate volatility and economic disruption.

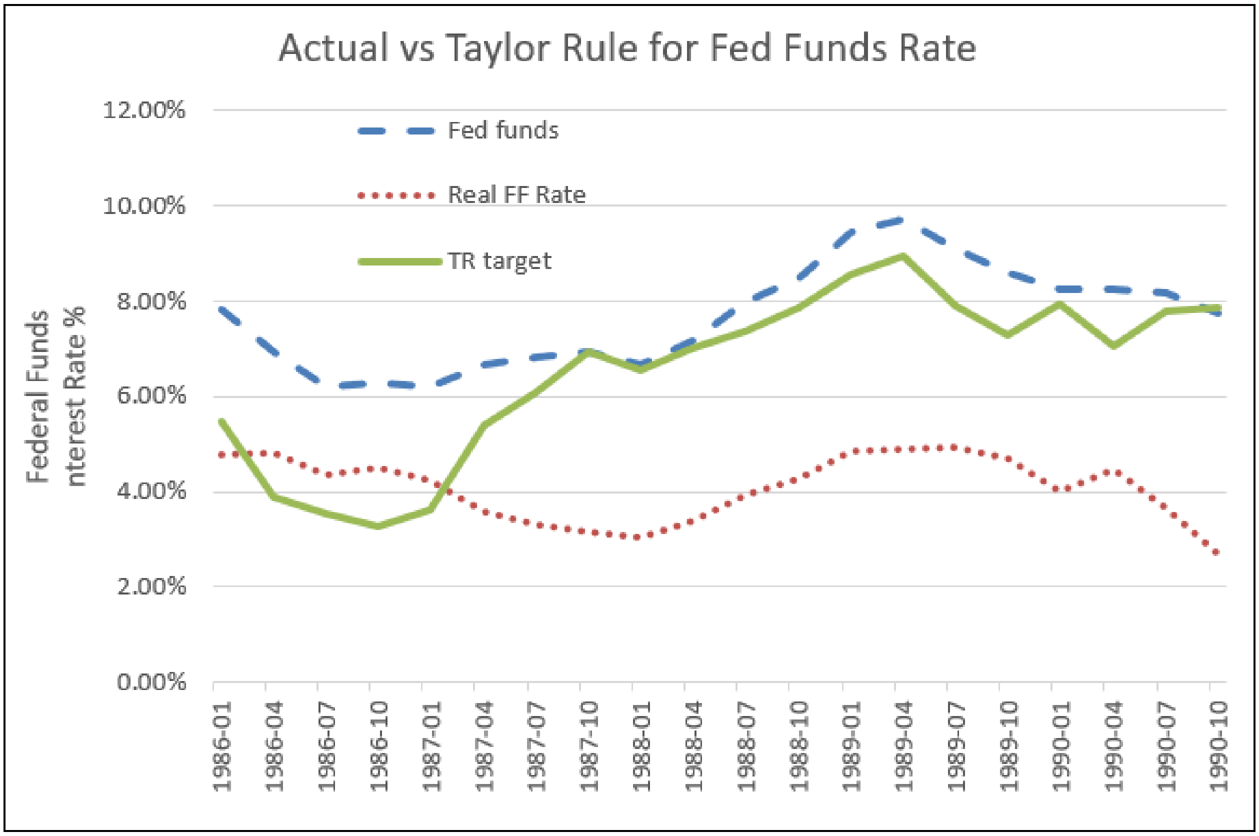
Economic Shocks and Inflationary Pressures

In early 1988, the U.S. economy experienced a positive real spending shock. The FOMC identified this as a robust business expansion occurring while industrial capacity was already near maximum. This spending shock was largely fueled by the government failing to withdraw their emergency efforts quickly after Black Monday. Despite the Federal Reserve's contractionary efforts to cool the economy, inflation emerged as a rising challenge, climbing steadily through the end of the decade.



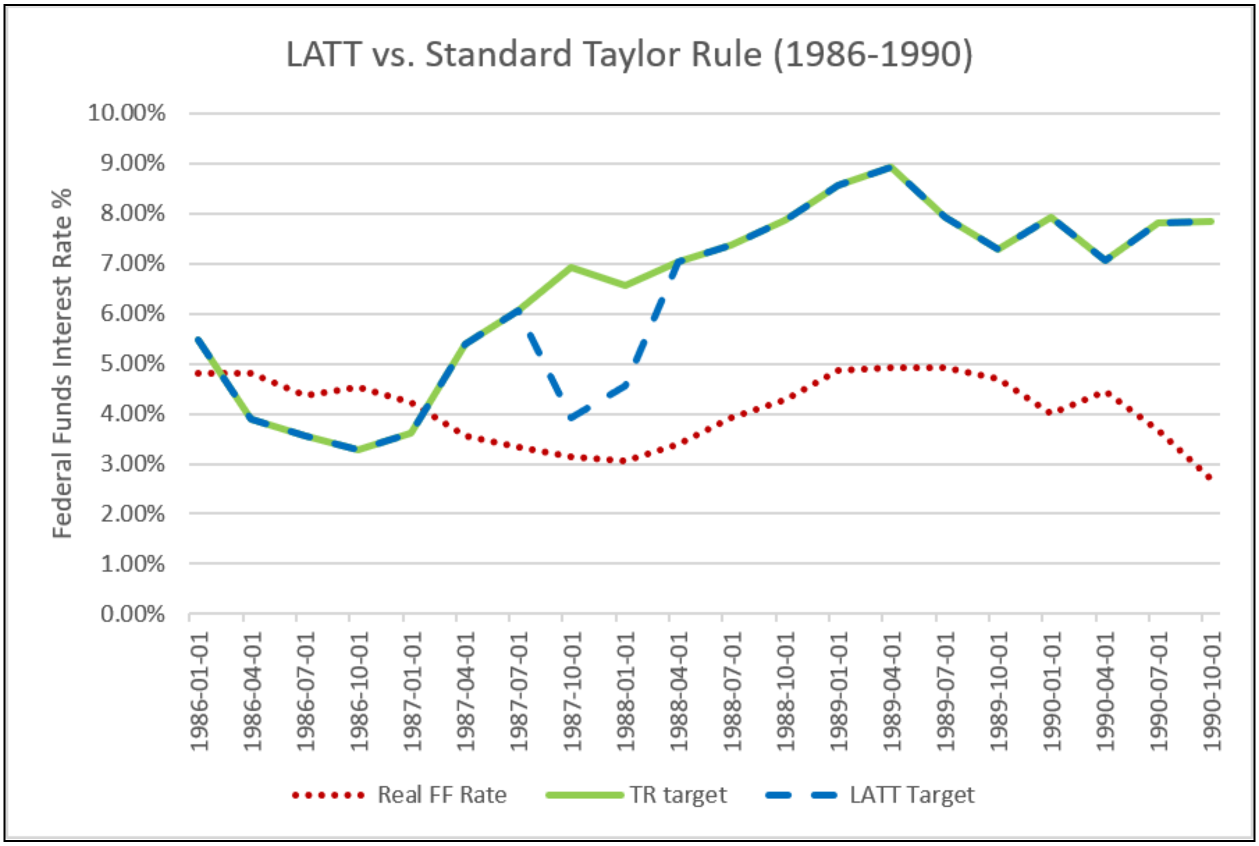
Policy Analysis and the Liquidity-Adjusted Taylor Target

The Federal Reserve's broad policy stance from 1986 to 1990 was contractionary. The actual federal funds rate remained consistently higher than the neutral real rate for the majority of the period. Furthermore, FOMC minutes explicitly document the decision to increase reserve pressures to combat the 1988 economic boom. Comparing the actual federal funds rate to the standard Taylor Rule target confirms this bias. The Fed maintained rates noticeable above the Taylor Rule benchmark for nearly the entire five year span.



However, the standard Taylor Rule alone has significant shortcomings. It is completely blind to financial crises and offers no guidance during severe market panics or banking collapses. To address this limitation within the 1986-1990 context, this analysis introduces the Liquidity-Adjusted Taylor Target (LATT).

This model adapts the standard Taylor Rule by incorporating a Financial Liquidity Override. During the October 1987 stock market crash (Black Monday), the standard Taylor Rule would have prescribed high interest rates due to rising inflation. This would have drained market liquidity and risked a severe depression. The LATT corrects this by automatically subtracting 3% from the target rate during a severe market panic and would inject emergency liquidity into the financial system.



As shown above, the LATT dips sharply in the late 1987 to stabilize the markets, before realigning with the standard Taylor Rule in mid-1988 once the panic subsided, allowing the Fed to resume its inflation targeting mandate.

Conclusion

A lesson from the 1986-1990 period is the necessity of flexibility within the central bank. The 1987 crash demonstrated that strictly following mathematical models like the Taylor Rule is insufficient. Central banks must occasionally prioritize market stabilization over short-term inflation targets to prevent banking failures. Furthermore, the modern Federal Reserve's commitment to transparency represents a significant improvement over the practices of the 1988 Fed. Today's approach of explicitly communicating interest rate targets helps anchor public expectations and mitigates unnecessary market volatility.

—---

Data Sources

—Board of Governors of the Federal Reserve System (1988). Minutes of the Federal Open Market Committee, March 29, 1988.

—Board of Governors of the Federal Reserve System (1988). Record of Policy Actions of the Federal Open Market Committee, March 29, 1988.

—Federal Reserve Economic Data (FRED): Consumer Price Index for All Urban Consumers: All Items in U.S. City Average (CPIAUCSL).

—Federal Reserve Economic Data (FRED): M1 and M2 Money Stock.

—Federal Reserve Economic Data (FRED): Effective Federal Funds Rate.